

Batch 2 – Module 2

Project 1

Project Title

Designing a Park Using Coordinate Geometry in Python

Project Description

In this project, students will apply concepts of **coordinate geometry** from the IGCSE Mathematics syllabus and implement them using **Python programming**. The goal is to design a simple park layout on a Cartesian plane and analyze its features mathematically through code.

The park is represented using coordinates, straight lines, and points. Key elements such as the park boundary, a walking path, benches, and a lamp post are plotted using Python's plotting library. Students will calculate distances, midpoints, gradients, and equations of straight lines, then verify their results visually on the graph.

This project helps students connect mathematical formulas with real-life representations and strengthens their understanding of coordinate geometry while introducing them to basic programming and data visualization. The level of difficulty is suitable for average IGCSE students and does not require advanced programming skills.

Project Objectives

By completing this project, students will be able to:

- Plot points and straight lines on a Cartesian plane using Python
- Represent real-world objects using coordinates
- Calculate the distance between two points
- Find the midpoint of a line segment
- Determine the gradient of a straight line
- Form the equation of a straight line
- Interpret graphical output to confirm mathematical results

Algorithm

1. Start the program
2. Import required Python libraries (`matplotlib`, `math`)
3. Define coordinates for the park boundary
4. Define coordinates for the walking path (two points)
5. Calculate the midpoint of the walking path
6. Define coordinates for benches
7. Plot the park boundary on the Cartesian plane
8. Plot the walking path
9. Plot the benches as points
10. Plot the midpoint as the lamp post
11. Add labels, grid, title, and legend to the graph
12. Display the graph
13. Calculate and display:
 - Distance of the walking path
 - Gradient of the walking path
 - Equation of the walking path
14. End the program

Pseudocode

START

IMPORT matplotlib

IMPORT math

SET boundary_x, boundary_y coordinates

SET path start point (x1, y1)

SET path end point (x2, y2)

CALCULATE midpoint_x = $(x1 + x2) / 2$

CALCULATE midpoint_y = $(y1 + y2) / 2$

SET bench coordinates

PLOT park boundary

PLOT walking path

PLOT benches as points

PLOT midpoint as lamp post

LABEL x-axis and y-axis

ADD title and legend

DISPLAY grid

SHOW graph

CALCULATE distance using distance formula

CALCULATE gradient using slope formula

CALCULATE equation of straight line

DISPLAY calculated results

END